

Carbon Price Volatility and Stock Returns

Aryan Goswami¹ and Sebastian Rink^{*1}

¹Frankfurt School of Finance & Management

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Abstract

We study whether carbon price uncertainty alters the pricing of firm carbon exposure. Using a monthly panel of STOXX Europe 600 firms with emission data and the option-implied Carbon VIX (CVIX), we show carbon return is state-dependent. High-emission firms show a modest discount in low-volatility months but a 10-point CVIX increase raises monthly excess returns for top emitters by ≈ 0.30 percentage points. Findings imply investors and policymakers should account for carbon price uncertainty, not just price levels, in asset pricing models and policy development explicitly.

JEL Codes: G12, Q51, Q54, Q58

Keywords: climate risk, carbon price volatility, carbon premium, asset pricing

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*Email: s.rink@fs.de

I. Introduction

Climate policy and the transition to a low-carbon economy affect corporate investment and financial markets (e.g., Acemoglu et al., 2018; Chava, 2014; Fuchs et al., 2024; Krueger et al., 2020; Starks, 2023). Regulatory instruments such as cap-and-trade systems convert greenhouse-gas (GHG) emissions into tradable costs that firms must internalize (Coase, 1960; Dales, 2002; Weitzman, 1974), and the rise of disclosure regimes makes these exposures increasingly observable (Krueger et al., 2024). Market participants must consider those developments from two perspectives: as the level of (future) carbon costs and as the uncertainty about their trajectory influenced by policy timing, permit supply, technology adoption, or geopolitical shocks. Hence, climate transition risk is an expected cost and a source of time-varying uncertainty that may alter required returns by investors.

This paper asks the question: does uncertainty in carbon markets change the way equity markets price firm-level carbon exposure? Specifically, we test whether firms with high emissions earn systematically different returns when the market's expectation of carbon-price volatility, measured by the option-implied Carbon VIX (CVIX) by Fuchs et al. (2024), is elevated. Our principal contribution is to show that the relation between emissions and stock returns is state-dependent: high-emission firms tend to trade at a modest discount in very tranquil but rare carbon price regimes, but earn statistically and economically meaningful higher returns when CVIX spikes.

Our empirical identification is based on a comprehensive monthly panel of STOXX Europe 600 constituents over 2013–2022, merged with firm-level GHG data from third-party providers, with EU allowance prices and the option-implied CVIX. We quantify carbon exposure in multiple ways and implement a panel regression where returns are regressed on emissions and the interaction of emissions with CVIX.

The results show that high-emission firms exhibit a negative return differential in low-volatility months, but the interaction with CVIX is positive and highly significant. Quantitatively, a 10-point rise in CVIX corresponds to roughly +0.30 percentage points of additional monthly excess return for top emitters, sufficient to turn a small monthly discount in tranquil and rare periods into a significant annualized excess return in high-volatility episodes. Scope-specific estimates indicate the effect is present for direct (Scope 1) and energy-related

(Scope 2) emissions and, somewhat attenuated, for value chain (Scope 3) emissions. The findings are robust to a battery of robustness tests.

The findings have several implications. Theoretically, they align with equilibrium asset-pricing intuition that investors demand higher compensation for risks whose uncertainty increases, reconciling apparently contradictory findings in the literature. Practically, they imply that static carbon screens or one-time assessments of climate exposure can misstate expected returns if they ignore volatility regimes. For policymakers, the results underscore the value of policy predictability when implementing climate policies.

The findings advance two strands of literature. First, we reconcile conflicting evidence regarding the existence of a carbon premium. While some studies document a positive premium compensating for transition risk (e.g., Bolton and Kacperczyk, 2021, 2023), others find a discount or insignificant results (Zhang, 2025). The literature attributes this heterogeneity to varying drivers, including measurement choices (Aswani et al., 2024), equity duration (He et al., 2024), or investor preferences (Pástor et al., 2021). We advance this debate by showing state dependency: we demonstrate that carbon price volatility acts as the regime-switching mechanism determining the sign and significance. Second, we extend the literature on policy uncertainty (e.g., Baker et al., 2016; Pástor and Veronesi, 2012, 2013). We isolate carbon market volatility as a distinct transmission channel, complementing recent findings on environmental policy uncertainty in finance (e.g., Hsu et al., 2023; Fuchs et al., 2024) by demonstrating how carbon price volatility affects equity returns.

The paper is structured as follows. In section II, we present our data and methodology. In section III, we discuss our findings. Section IV concludes.

II. Data and Methodology

A. Data

We assemble a monthly panel of STOXX Europe 600 firms from 2013–2022. The sample comprises all firms that were STOXX Europe 600 constituents as of December 2022. Carbon data are obtained from S&P GLOBAL TRUCOST, which reports firm-level greenhouse gas (GHG) emissions and carbon intensity. We use one-year lagged Scope 1 (direct), Scope 2 (energy-

related), and Scope 3 (value chain) emissions, in both absolute (tons of CO₂-equivalent) and revenue-normalized (tons of CO₂-equivalent per thousand euros) forms. Stock returns come from COMPUSTAT GLOBAL SECURITY DAILY (via WRDS) and are aggregated from daily closing prices to monthly percentage returns. We construct the return-based controls momentum, defined as the cumulative return over the prior 12 months excluding the most recent month, and idiosyncratic volatility, the annualized standard deviation of daily returns over a 252-day rolling window. Annual fundamentals following Bolton and Kacperczyk (2021) are taken from COMPUSTAT GLOBAL FUNDAMENTALS ANNUAL, forward filled to monthly frequency and lagged by six months relative to fiscal year-end, following standard asset pricing practice (Fama and French, 1992, 1993). We retain firm-months observations with non-missing returns, fundamentals, and emissions data, leading to the final sample in Table I. Market-level carbon conditions are captured using European Union Allowance (EUA) prices and the CARBONVIX (CVIX) index from carbonvix.org by Fuchs et al. (2024). All continuous variables are winsorized at the 2.5% levels.

B. Empirical Setting

Our main specification is:

$$r_{i,t} = \alpha + \beta_1 \text{HighEmit}_{i,t} + \beta_2 (\text{HighEmit}_{i,t} \times \text{CVIX}_t) + \gamma' \mathbf{X}_{i,t} + \lambda_{j(i)} + \mu_t + \varepsilon_{i,t}, \quad (1)$$

where $r_{i,t}$ is the monthly return of firm i in month t ; $\text{HighEmit}_{i,t}$ equals one if firm i is in the top 20% of emitters; CVIX_t is the monthly carbon volatility index; $\mathbf{X}_{i,t}$ is the control vector; $\lambda_{j(i)}$ are industry fixed effects; and μ_t are year-month fixed effects.

Here, β_1 captures the unconditional carbon return while β_2 measures how this return varies with carbon volatility. A positive β_2 implies that high-emitting firms earn higher returns in periods of elevated carbon market uncertainty.

Equation (1) is estimated via panel OLS with two-way clustered standard errors (by industry and month) following Petersen (2009). Industry fixed effects control for persistent sectoral differences, and time effects absorb macro shocks. Since firm-level emissions change slowly, industry rather than firm fixed effects preserve sufficient variation to identify β_1 and β_2 .

We assess robustness through several extensions. First, we replace the high-emitter dummy

Variable	Mean	Std. Dev.	P25	Median	P75
Monthly Excess Return (%)	0.0070	0.0718	-0.0367	0.0078	0.0518
Carbon VIX (Monthly)	54.4260	9.9756	47.6098	54.1453	59.5044
log(Scope 1)	11.2935	2.9786	9.2954	11.0971	12.9775
log(Scope 2)	11.3502	2.1524	9.9846	11.4185	12.7434
log(Scope 3)	13.5771	2.1166	12.0866	13.6794	15.1590
Intensity Scope 1	0.1473	0.4503	0.0027	0.0130	0.0513
Intensity Scope 2	0.0469	0.1639	0.0057	0.0152	0.0344
Intensity Scope 3	0.2965	1.5278	0.0470	0.1260	0.3011
Log Size	9.5431	1.2548	8.6069	9.4221	10.4724
Book-to-Market	0.6262	0.5662	0.2389	0.4399	0.8304
Momentum (%)	0.0793	0.2796	-0.0981	0.0644	0.2384
Leverage	0.2421	0.1529	0.1299	0.2301	0.3376
CapEx / Total Assets	0.0380	0.0266	0.0180	0.0323	0.0519
Log PPE	7.5010	2.0880	6.1860	7.7066	9.0445
Sales Growth (%)	0.0001	0.0277	0.0000	0.0000	0.0054
EPS Growth (%)	0.0039	0.0938	-0.0095	0.0058	0.0259
Return on Equity (%)	0.1305	0.1295	0.0657	0.1195	0.1856
Idiosyncratic Volatility (1Y)	0.2778	0.1067	0.2026	0.2516	0.3229
Market Beta	0.9556	0.3515	0.7211	0.9208	1.1597
PPE Intensity	0.2076	0.2079	0.0345	0.1375	0.3151

Table I. Summary Statistics

with continuous measures, such as $\log(1 + \text{Scope}_i)$ or emissions intensity. Second, we vary the fixed-effects structure, re-estimating with firm fixed effects. Third, we test stability by excluding the 2020–2021 COVID-19 period.

III. Results and Discussion

Tables II and III report our core results using a top-20% high-emitter dummy under absolute emissions and emission intensity respectively. The high-emitter dummy is negative and statistically significant at low levels of carbon-market volatility, indicating a carbon discount in tranquil but rare regimes. The interaction between the high-emitter indicator and the CVIX is positive and highly significant in all specifications, implying that the discount attenuates and eventually reverses as carbon-market volatility increases. Evaluated at observed CVIX values, the implied break-even lies consistently in the $\text{CVIX} \approx 40\text{--}50$ range. Above

this threshold, high emitters earn an excess return relative to lower-emission peers.

Quantitatively, a 10-point increase in CVIX is associated with roughly a +0.30 percentage-point increase in monthly excess return for top emitters, all other things equal. This state-dependent reversal is visible in extreme episodes. Low-CVIX months ($\text{CVIX} \sim 30$) imply a small monthly discount for high emitters ($\approx -0.01\%$ per month), whereas high-CVIX episodes ($\text{CVIX} \sim 80$) correspond to large positive monthly premiums ($\approx +1.4\text{--}1.6\%$ per month). Figure 1 plots the fitted high-emitter return as a function of CVIX (holding covariates at sample means), showing the sign flip and 95% confidence bands.

Variables	No Industry FE			With Industry FE		
	Scope 1	Scope 2	Scope 3	Scope 1	Scope 2	Scope 3
Intercept	0.0309*** (0.0042)	0.0303*** (0.0042)	0.0301*** (0.0042)	0.0480*** (0.0059)	0.0491*** (0.0059)	0.0503*** (0.0060)
Dummy (top 20%)	-0.0117*** (0.0042)	-0.0164*** (0.0042)	-0.0180*** (0.0041)	-0.0113** (0.0045)	-0.0169*** (0.0043)	-0.0202*** (0.0043)
Dummy $\times$ CVIX	0.0003*** (0.0001)	0.0003*** (0.0001)	0.0003*** (0.0001)	0.0003*** (0.0001)	0.0003*** (0.0001)	0.0003*** (0.0001)
Momentum	-0.0025 (0.0019)	-0.0025 (0.0019)	-0.0026 (0.0019)	-0.0090*** (0.0021)	-0.0089*** (0.0021)	-0.0091*** (0.0021)
Book-to-Market	-0.0175*** (0.0021)	-0.0171*** (0.0020)	-0.0168*** (0.0020)	-0.0220*** (0.0022)	-0.0217*** (0.0023)	-0.0216*** (0.0023)
Leverage	-0.0101*** (0.0029)	-0.0100*** (0.0029)	-0.0101*** (0.0029)	-0.0115*** (0.0043)	-0.0115*** (0.0043)	-0.0120*** (0.0043)
CapEx / TA	-0.0909*** (0.0199)	-0.0928*** (0.0198)	-0.0913*** (0.0199)	-0.1285*** (0.0261)	-0.1271*** (0.0258)	-0.1278*** (0.0258)
PPE intensity	0.0119*** (0.0032)	0.0134*** (0.0032)	0.0137*** (0.0032)	0.0117** (0.0053)	0.0109** (0.0052)	0.0100* (0.0052)
Sales growth	0.0152 (0.0138)	0.0142 (0.0138)	0.0130 (0.0137)	0.0090 (0.0128)	0.0087 (0.0129)	0.0080 (0.0128)
EPS growth	0.0070* (0.0037)	0.0072* (0.0038)	0.0068* (0.0038)	0.0086** (0.0039)	0.0088** (0.0038)	0.0081** (0.0039)
Return on Equity	-0.0175*** (0.0038)	-0.0177*** (0.0037)	-0.0179*** (0.0037)	-0.0236*** (0.0046)	-0.0233*** (0.0045)	-0.0232*** (0.0045)
Volatility (1Y)	0.0230*** (0.0052)	0.0222*** (0.0053)	0.0212*** (0.0052)	0.0149** (0.0067)	0.0140** (0.0068)	0.0133** (0.0068)
Market Beta	0.0020 (0.0016)	0.0025 (0.0016)	0.0029* (0.0016)	0.0001 (0.0020)	0.0007 (0.0019)	0.0014 (0.0020)
R-squared	0.285	0.285	0.285	0.291	0.291	0.291
Observations	34186	34186	34186	34186	34186	34186
Industry FE	No	No	No	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The dependent variable is monthly excess return. “Dummy (top 20%)” equals 1 for firms in the top 20% of absolute emissions within the relevant scope (Scope 1, 2, or 3) in each cross-section. Columns (1)–(3) exclude industry fixed effects; (4)–(6) include industry fixed effects. Controls: momentum, book-to-market, leverage, CapEx/total assets, PPE intensity, sales growth, EPS growth, ROE, 1-year return volatility, and market beta. Time fixed effects are included in all columns. Two-way clustered standard errors in parentheses. Sample: 34,186 firm-months. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table II. Excess Return Using 20% Emission Dummy - Absolute
Monthly excess returns regressed on a high-emitter dummy (top 20% emissions, scope-specific) and its interaction with the Carbon Volatility Index (CVIX).

Variables	No Industry FE			With Industry FE		
	Scope 1	Scope 2	Scope 3	Scope 1	Scope 2	Scope 3
Intercept	0.0309*** (0.0042)	0.0308*** (0.0042)	0.0306*** (0.0042)	0.0459*** (0.0061)	0.0483*** (0.0058)	0.0495*** (0.0059)
Dummy (top 20%)	-0.0144*** (0.0051)	-0.0128** (0.0055)	-0.0113** (0.0045)	-0.0141*** (0.0052)	-0.0115** (0.0058)	-0.0106** (0.0048)
Dummy $\times$ CVIX	0.0003*** (0.0001)	0.0003** (0.0001)	0.0002** (0.0001)	0.0003*** (0.0001)	0.0003** (0.0001)	0.0002** (0.0001)
Momentum	-0.0027 (0.0019)	-0.0024 (0.0019)	-0.0024 (0.0019)	-0.0091*** (0.0021)	-0.0089*** (0.0021)	-0.0088*** (0.0021)
Book-to-Market	-0.0175*** (0.0022)	-0.0170*** (0.0020)	-0.0170*** (0.0020)	-0.0219*** (0.0023)	-0.0217*** (0.0023)	-0.0217*** (0.0023)
Leverage	-0.0097*** (0.0029)	-0.0099*** (0.0029)	-0.0101*** (0.0029)	-0.0108** (0.0043)	-0.0117*** (0.0043)	-0.0116*** (0.0043)
CapEx / TA	-0.0897*** (0.0198)	-0.0916*** (0.0199)	-0.0907*** (0.0201)	-0.1288*** (0.0261)	-0.1304*** (0.0258)	-0.1268*** (0.0258)
PPE intensity	0.0102*** (0.0031)	0.0125*** (0.0033)	0.0135*** (0.0032)	0.0113** (0.0052)	0.0100* (0.0053)	0.0109** (0.0052)
Sales growth	0.0165 (0.0133)	0.0146 (0.0139)	0.0145 (0.0139)	0.0103 (0.0124)	0.0096 (0.0130)	0.0094 (0.0130)
EPS growth	0.0070* (0.0037)	0.0072* (0.0037)	0.0072* (0.0038)	0.0086** (0.0039)	0.0086** (0.0038)	0.0086** (0.0039)
Return on Equity	-0.0174*** (0.0038)	-0.0177*** (0.0037)	-0.0179*** (0.0037)	-0.0234*** (0.0046)	-0.0234*** (0.0045)	-0.0233*** (0.0045)
Volatility (1Y)	0.0219*** (0.0051)	0.0216*** (0.0051)	0.0218*** (0.0051)	0.0139** (0.0067)	0.0139** (0.0068)	0.0142** (0.0067)
Market Beta	0.0021 (0.0016)	0.0024 (0.0015)	0.0026* (0.0016)	0.0005 (0.0019)	0.0005 (0.0019)	0.0006 (0.0019)
R-squared	0.285	0.285	0.285	0.291	0.291	0.291
Observations	34186	34186	34186	34186	34186	34186
Industry FE	No	No	No	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Columns (1)–(3) exclude industry fixed effects; (4)–(6) include industry fixed effects. “Dummy (top 20%)” equals 1 for firms in the top 20% of relative emissions (tons of CO₂-equivalent per thousand euros) within the relevant scope (Scope 1, 2, or 3) in each cross-section. Controls: momentum, book-to-market, leverage, CapEx/total assets, PPE intensity, sales growth, EPS growth, ROE, 1-year return volatility, and market beta. Time fixed effects are included in all columns. Two-way clustered standard errors in parentheses. Sample: 34,186 firm-months.

Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table III. Excess Return Using 20% Emission Dummy - Intensity

Monthly excess returns regressed on a high-emitter dummy (top 20% relative emissions, scope-specific) and its interaction with the Carbon Volatility Index (CVIX).

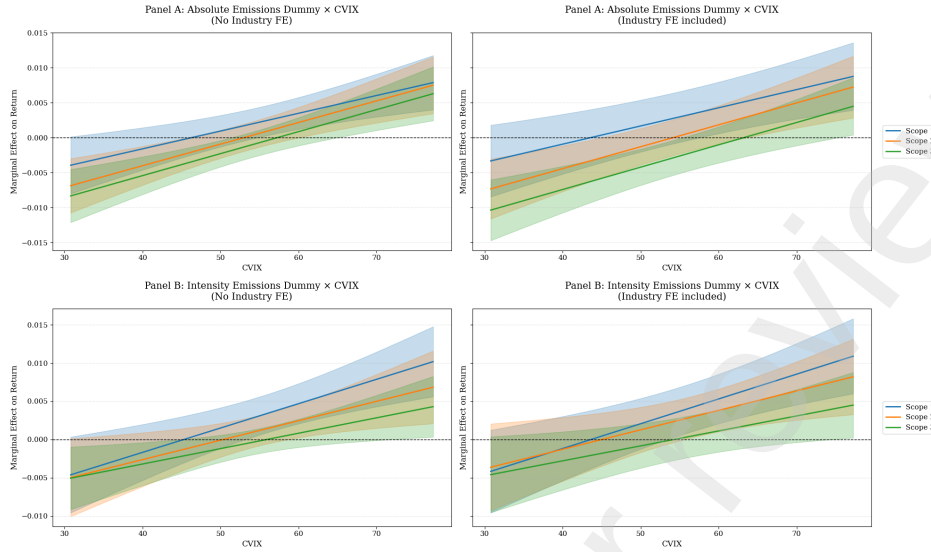


Figure 1. Predicted returns for high emitters across CVIX

In terms of different emission scopes, the unconditional negative discount ($CVIX=0$) is somewhat higher for Scope 2 and Scope 3 in absolute terms, while Scope 1 shows a lower baseline effect. However, the interaction coefficients across scopes are nearly identical ($\approx +0.0003$ per CVIX point), indicating that volatility mitigates carbon discounts broadly. Scope 3 estimates are less precisely estimated than Scope 1/2, consistent with a higher measurement error and reporting heterogeneity in value chain emissions (Busch et al., 2022).

The empirical pattern indicates a *state-contingent* pricing of carbon transition risk. When CVIX is low (stable carbon prices and limited policy uncertainty) high-emission firms on average trade at a discount or at least not at higher returns. This likely reflects emission-driven capital reallocation, longer-term regulatory concerns priced into valuations, or persistent investor preferences that penalize carbon-intensive exposures. By contrast, when CVIX is elevated and uncertainty about carbon prices and policy intensifies, high emissions firm earn higher contemporaneous returns. The observed CVIX threshold ($\approx 40-50$) can be interpreted as a tipping point at which the transition risk moves from a latent concern to a salient state variable that is capitalized into prices.

Replacing the top-20% dummy with continuous log-emissions yields the same conclusion: absolute emissions have a negative unconditional coefficient and a positive interaction with

CVIX. Intensity-based measures produce qualitatively similar but weaker interaction effects. That pattern is consistent with absolute emissions being a more direct proxy for potential regulatory liabilities and aggregate carbon costs; intensity measures introduce additional noise from firm size and sectoral structure. Results using continuous emissions are summarized in Tables IV & V and Figure 2.

Variables	No Industry FE			With Industry FE		
	Scope 1	Scope 2	Scope 3	Scope 1	Scope 2	Scope 3
Intercept	0.0164*** (0.0050)	0.0165*** (0.0064)	0.0097 (0.0065)	0.0349*** (0.0075)	0.0366*** (0.0083)	0.0289*** (0.0085)
Absolute Emissions (log)	-0.0036*** (0.0007)	-0.0053*** (0.0009)	-0.0054*** (0.0010)	-0.0037*** (0.0007)	-0.0054*** (0.0010)	-0.0055*** (0.0011)
Emissions $\times$ CVIX	0.0001*** (0.0000)	0.0001*** (0.0000)	0.0001*** (0.0000)	0.0001*** (0.0000)	0.0001*** (0.0000)	0.0001*** (0.0000)
Momentum	-0.0026 (0.0019)	-0.0028 (0.0019)	-0.0026 (0.0019)	-0.0091*** (0.0022)	-0.0092*** (0.0022)	-0.0090*** (0.0022)
Book-to-Market	-0.0173*** (0.0022)	-0.0166*** (0.0020)	-0.0169*** (0.0020)	-0.0218*** (0.0023)	-0.0214*** (0.0023)	-0.0216*** (0.0023)
Leverage	-0.0101*** (0.0029)	-0.0096*** (0.0029)	-0.0100*** (0.0029)	-0.0117*** (0.0043)	-0.0111*** (0.0042)	-0.0118*** (0.0043)
CapEx / TA	-0.0919*** (0.0201)	-0.0871*** (0.0199)	-0.0912*** (0.0201)	-0.1274*** (0.0265)	-0.1249*** (0.0257)	-0.1269*** (0.0262)
PPE intensity	0.0128*** (0.0032)	0.0138*** (0.0032)	0.0135*** (0.0032)	0.0109** (0.0053)	0.0107** (0.0051)	0.0107** (0.0053)
Sales growth	0.0145 (0.0136)	0.0120 (0.0139)	0.0136 (0.0137)	0.0086 (0.0126)	0.0070 (0.0132)	0.0085 (0.0128)
EPS growth	0.0067* (0.0038)	0.0071* (0.0038)	0.0066* (0.0038)	0.0082** (0.0039)	0.0084** (0.0038)	0.0079** (0.0039)
Return on Equity	-0.0177*** (0.0037)	-0.0184*** (0.0037)	-0.0180*** (0.0038)	-0.0234*** (0.0046)	-0.0237*** (0.0045)	-0.0235*** (0.0046)
Volatility (1Y)	0.0228*** (0.0056)	0.0198*** (0.0057)	0.0218*** (0.0055)	0.0146** (0.0069)	0.0128* (0.0070)	0.0143** (0.0070)
Market Beta	0.0023 (0.0016)	0.0032** (0.0016)	0.0027* (0.0016)	0.0004 (0.0019)	0.0011 (0.0020)	0.0008 (0.0020)
R-squared	0.285	0.285	0.285	0.291	0.291	0.291
Observations	34186	34186	34186	34186	34186	34186
Industry FE	No	No	No	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The dependent variable is monthly excess return. Columns (1)–(3) exclude industry fixed effects; (4)–(6) include industry fixed effects. Controls: momentum, book-to-market, leverage, CapEx/total assets, PPE intensity, sales growth, EPS growth, ROE, 1-year return volatility, and market beta. Time fixed effects are included in all columns. Two-way clustered standard errors are in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table IV. Excess Return Continuous Absolute Emissions

Monthly excess returns regressed on firm-level log absolute emissions and their interaction with the Carbon Volatility Index (CVIX).

Variables	No Industry FE			With Industry FE		
	Scope 1	Scope 2	Scope 3	Scope 1	Scope 2	Scope 3
Intercept	0.0310*** (0.0042)	0.0310*** (0.0042)	0.0307*** (0.0042)	0.0500*** (0.0059)	0.0500*** (0.0059)	0.0489*** (0.0058)
Emissions Intensity	-0.0118*** (0.0030)	-0.0307*** (0.0084)	-0.0103* (0.0057)	-0.0133*** (0.0030)	-0.0320*** (0.0089)	-0.0105* (0.0058)
Emissions $\times$ CVIX	0.0002*** (0.0001)	0.0006*** (0.0002)	0.0002** (0.0001)	0.0002*** (0.0001)	0.0006*** (0.0002)	0.0002** (0.0001)
Momentum	-0.0025 (0.0019)	-0.0025 (0.0019)	-0.0025 (0.0019)	-0.0090*** (0.0021)	-0.0089*** (0.0021)	-0.0090*** (0.0021)
Book-to-Market	-0.0171*** (0.0021)	-0.0171*** (0.0020)	-0.0172*** (0.0020)	-0.0217*** (0.0023)	-0.0217*** (0.0023)	-0.0219*** (0.0023)
Leverage	-0.0101*** (0.0029)	-0.0098*** (0.0029)	-0.0094*** (0.0029)	-0.0117*** (0.0043)	-0.0113*** (0.0043)	-0.0112*** (0.0043)
CapEx / TA	-0.0896*** (0.0200)	-0.0907*** (0.0200)	-0.0912*** (0.0202)	-0.1264*** (0.0259)	-0.1252*** (0.0259)	-0.1255*** (0.0257)
PPE intensity	0.0127*** (0.0032)	0.0135*** (0.0032)	0.0132*** (0.0032)	0.0107** (0.0052)	0.0110** (0.0053)	0.0111** (0.0052)
Sales growth	0.0141 (0.0136)	0.0148 (0.0138)	0.0150 (0.0136)	0.0084 (0.0126)	0.0096 (0.0129)	0.0095 (0.0127)
EPS growth	0.0070* (0.0038)	0.0075** (0.0038)	0.0072* (0.0038)	0.0087** (0.0039)	0.0089** (0.0039)	0.0086** (0.0038)
Return on Equity	-0.0176*** (0.0037)	-0.0184*** (0.0036)	-0.0179*** (0.0037)	-0.0232*** (0.0045)	-0.0241*** (0.0044)	-0.0238*** (0.0045)
Volatility (1Y)	0.0216*** (0.0050)	0.0216*** (0.0051)	0.0219*** (0.0051)	0.0137** (0.0067)	0.0138** (0.0067)	0.0138** (0.0067)
Market Beta	0.0024 (0.0016)	0.0027* (0.0015)	0.0024 (0.0015)	0.0006 (0.0019)	0.0007 (0.0019)	0.0006 (0.0019)
R-squared	0.285	0.285	0.285	0.291	0.291	0.291
Observations	34186	34186	34186	34186	34186	34186
Industry FE	No	No	No	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The dependent variable is monthly excess return. Columns (1)–(3) exclude industry fixed effects; (4)–(6) include them. Emission intensities are in tons of CO₂-equivalent per thousand euros. Controls: momentum, book-to-market, leverage, CapEx/total assets, PPE intensity, sales growth, EPS growth, ROE, 1-year return volatility, and market beta. Time fixed effects are included in all columns. Controls and sample are identical to those in Table IV. Two-way clustered standard errors in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table V. Excess Return Continuous Emission Intensities

Monthly excess returns regressed on firm-level emissions intensity and their interaction with the Carbon Volatility Index (CVIX).

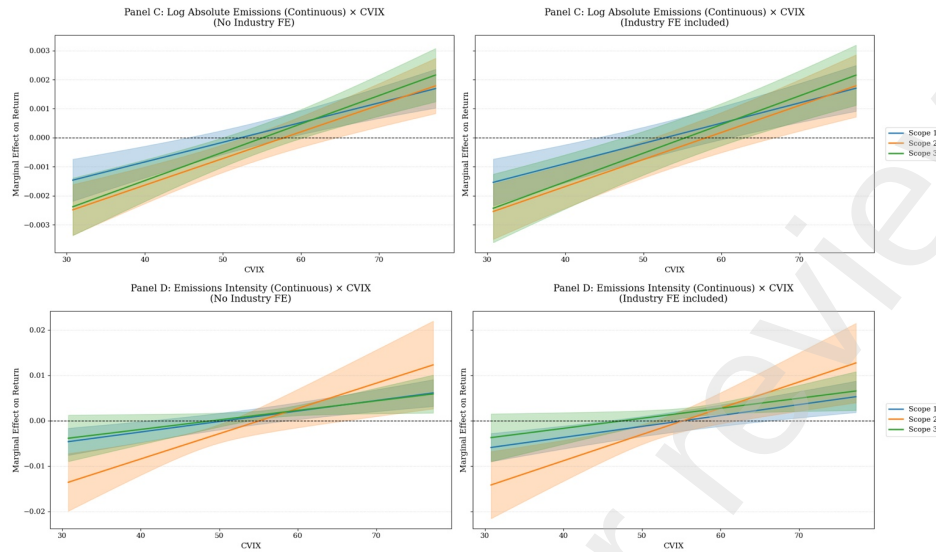


Figure 2. Predicted returns for continuous emissions across CVIX

We conduct further robustness checks. Key findings are unchanged when excluding the COVID-19 pandemic period (2020–21) and substituting industry fixed effects for firm fixed effects. See Appendix A for regression results.

IV. Conclusion

This paper shows that the pricing of carbon transition risk in equity markets depends on the volatility regime in carbon markets. Two implications follow. First, carbon transition risk behaves as a time-varying priced factor: static screens or cross-sectional carbon scores may mischaracterize expected returns unless they account for volatility regimes. Second, policy uncertainty that elevates CVIX can trigger abrupt repricing, leading to adjustments in valuations and return differentials across firms.

V. AI Declaration

During the preparation of this manuscript, the authors used Gemini and ChatGPT to improve language readability and formatting. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

References

- Acemoglu, D., Akcigit, U., Alp, H., Bloom, N., Kerr, W., 2018. Innovation, Reallocation, and Growth. *American Economic Review* 108, 3450–3491. doi:10.1257/aer.20130470.
- Aswani, J., Raghunandan, A., Rajgopal, S., 2024. Are Carbon Emissions Associated with Stock Returns? *Review of Finance* 28, 75–106. doi:10.1093/rof/rfad013.
- Baker, S.R., Bloom, N., Davis, S.J., 2016. Measuring Economic Policy Uncertainty. *The Quarterly Journal of Economics* 131, 1593–1636. doi:10.1093/qje/qjw024.
- Bolton, P., Kacperczyk, M., 2021. Do investors care about carbon risk? *Journal of Financial Economics* 142. doi:10.1016/j.jfineco.2021.05.008.
- Bolton, P., Kacperczyk, M., 2023. Global Pricing of Carbon-Transition Risk. *The Journal of Finance* 78, 3677–3754. doi:10.1111/jofi.13272.
- Busch, T., Johnson, M., Pioch, T., 2022. Corporate carbon performance data: Quo vadis? *Journal of Industrial Ecology* 26. doi:10.1111/jiec.13008.
- Chava, S., 2014. Environmental externalities and cost of capital. *Management Science* 60. doi:10.1287/mnsc.2013.1863.
- Coase, R.H., 1960. The Problem of Social Cost. *The Journal of Law and Economics* 3, 1–44. doi:10.1086/466560.
- Dales, J.H., 2002. *Pollution, property & prices : an essay in policy-making and economics*. Edward Elgar Pub.
- Fama, E.F., French, K.R., 1992. The Cross-Section of Expected Stock Returns. *The Journal of Finance* 47, 427–465. doi:10.1111/j.1540-6261.1992.tb04398.x.
- Fama, E.F., French, K.R., 1993. Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics* 33, 3–56. doi:10.1016/0304-405X(93)90023-5.
- Fuchs, M., Stroebel, J., Terstegge, J., 2024. Carbon VIX: Carbon Price Uncertainty and Decarbonization Investments
doi:10.2139/ssrn.4913715.

- He, Y., Li, J., Zhang, R., 2024. Duration-driven Carbon Premium. doi:10.2139/ssrn.4880807.
- Hsu, P., Li, K., Tsou, C., 2023. The Pollution Premium. *The Journal of Finance* 78, 1343–1392. doi:10.1111/jofi.13217.
- Krueger, P., Sautner, Z., Starks, L.T., 2020. The importance of climate risks for institutional investors. *Review of Financial Studies* 33. doi:10.1093/rfs/hhz137.
- Krueger, P., Sautner, Z., Tang, D.Y., Zhong, R., 2024. The Effects of Mandatory ESG Disclosure Around the World. *Journal of Accounting Research* doi:10.1111/1475-679X.12548.
- Pástor, L., Stambaugh, R.F., Taylor, L.A., 2021. Sustainable investing in equilibrium. *Journal of Financial Economics* 142. doi:10.1016/j.jfineco.2020.12.011.
- Pástor, L., Veronesi, P., 2012. Uncertainty about Government Policy and Stock Prices. *The Journal of Finance* 67, 1219–1264. doi:10.1111/j.1540-6261.2012.01746.x.
- Pástor, L., Veronesi, P., 2013. Political uncertainty and risk premia. *Journal of Financial Economics* 110, 520–545. URL: <https://www.sciencedirect.com/science/article/pii/S0304405X13002080>, doi:10.1016/J.JFINECO.2013.08.007.
- Petersen, M.A., 2009. Estimating standard errors in finance panel data sets: Comparing approaches. doi:10.1093/rfs/hhn053.
- Starks, L.T., 2023. Presidential Address: Sustainable Finance and ESG Issues—Value versus Values. *Journal of Finance* 78, 1837–1872. doi:10.1111/jofi.13255.
- Weitzman, M.L., 1974. Prices vs. Quantities. *The Review of Economic Studies* 41, 477. doi:10.2307/2296698.
- Zhang, S., 2025. Carbon Returns across the Globe. *The Journal of Finance* 80, 615–645. doi:10.1111/jofi.13402.

A. Appendix A

Table VI. Variable Definitions

Variable	Definition or construction	Source	Unit	Frequency
Monthly excess return	Monthly stock return of firm i in excess of the one month risk free rate	Compustat Global	Decimal	Monthly
Carbon VIX	Option implied volatility index of European Union Allowance prices	carbonvix.org	index points	Monthly
Absolute emissions Scope 1	Direct greenhouse gas emissions of the firm, lagged by one year	Trucost	Tons CO ₂ e	Annual, mapped to months
Absolute emissions Scope 2	Indirect emissions from purchased electricity and heat, lagged by one year	Trucost	Tons CO ₂ e	Annual, mapped to months
Absolute emissions Scope 3	Value chain emissions (upstream and downstream), lagged by one year	Trucost	Tons CO ₂ e	Annual, mapped to months
Emissions intensity Scope 1	Scope 1 emissions divided by firm revenue, scaled from tons per million euro to tons per euro	Trucost	Scaled tons/euro	Annual, mapped to months
Emissions intensity Scope 2	Scope 2 emissions divided by firm revenue, scaled from tons per million euro to tons per euro	Trucost	Scaled tons/euro	Annual, mapped to months
Emissions intensity Scope 3	Scope 3 emissions divided by firm revenue, scaled from tons per million euro to tons per euro	Trucost	Scaled tons/euro	Annual, mapped to months
Log size	Natural log of market capitalisation at month end	Compustat Global	Log euro	Monthly
Book to market ratio	Book value of equity divided by market value of equity	Compustat Global	Ratio	Monthly
Momentum	Cumulative stock return from month $t - 12$ to $t - 2$	Compustat Global	Decimal	Monthly
Leverage	Total debt divided by total assets	Compustat Global	Ratio	Monthly
Capital expenditure over total assets	Capital expenditures divided by total assets	Compustat Global	Ratio	Annual, carried forward

continued on next page

Table VI – continued from previous page

Variable	Definition or construction	Source	Unit	Frequency
Log PPE	Natural log of property, plant and equipment	Compustat Global	Log euro	Annual, carried forward
Sales growth	Growth in firm sales over the previous year	Compustat Global	Decimal	Annual, carried forward
EPS growth	Growth in earnings per share over the previous year	Compustat Global	Decimal	Annual, carried forward
Return on equity	Net income divided by book equity	Compustat Global	Decimal	Annual, carried forward
Idiosyncratic volatility	Annualised standard deviation of daily residuals from a market model over the past year	Compustat Global	Decimal	Monthly
Market beta	Beta with respect to the STOXX Europe 600 index estimated using daily returns	Compustat Global	Beta	Monthly
PPE intensity	Logarithm of the ratio of PPE to total assets	Compustat Global	Log ratio	Annual, carried forward

Variables	No Industry FE			With Industry FE		
	Scope 1	Scope 2	Scope 3	Scope 1	Scope 2	Scope 3
Intercept	0.0287*** (0.0040)	0.0279*** (0.0041)	0.0278*** (0.0040)	0.0475*** (0.0063)	0.0484*** (0.0063)	0.0497*** (0.0064)
Dummy (top 20%)	-0.0124*** (0.0044)	-0.0194*** (0.0043)	-0.0200*** (0.0043)	-0.0125*** (0.0047)	-0.0202*** (0.0044)	-0.0231*** (0.0045)
Dummy $\times$ CVIX	0.0003*** (0.0001)	0.0004*** (0.0001)	0.0003*** (0.0001)	0.0003*** (0.0001)	0.0004*** (0.0001)	0.0004*** (0.0001)
Momentum	-0.0018 (0.0021)	-0.0018 (0.0020)	-0.0018 (0.0020)	-0.0083*** (0.0024)	-0.0083*** (0.0023)	-0.0085*** (0.0024)
Book-to-Market	-0.0159*** (0.0021)	-0.0154*** (0.0021)	-0.0152*** (0.0020)	-0.0200*** (0.0022)	-0.0197*** (0.0023)	-0.0197*** (0.0023)
Leverage	-0.0085*** (0.0030)	-0.0084*** (0.0030)	-0.0084*** (0.0030)	-0.0124*** (0.0043)	-0.0125*** (0.0043)	-0.0130*** (0.0042)
CapEx / TA	-0.0892*** (0.0213)	-0.0895*** (0.0215)	-0.0892*** (0.0213)	-0.1374*** (0.0288)	-0.1342*** (0.0285)	-0.1354*** (0.0283)
PPE intensity	0.0102*** (0.0033)	0.0117*** (0.0032)	0.0119*** (0.0032)	0.0072 (0.0052)	0.0064 (0.0051)	0.0053 (0.0051)
Sales growth	0.0002 (0.0124)	-0.0010 (0.0122)	-0.0018 (0.0121)	-0.0036 (0.0117)	-0.0042 (0.0117)	-0.0050 (0.0115)
EPS growth	0.0099** (0.0044)	0.0103** (0.0044)	0.0097** (0.0044)	0.0113** (0.0047)	0.0116** (0.0046)	0.0111** (0.0047)
Return on Equity	-0.0201*** (0.0039)	-0.0204*** (0.0038)	-0.0203*** (0.0039)	-0.0242*** (0.0047)	-0.0240*** (0.0046)	-0.0238*** (0.0046)
Volatility (1Y)	0.0226*** (0.0056)	0.0215*** (0.0057)	0.0212*** (0.0056)	0.0150** (0.0074)	0.0142* (0.0075)	0.0136* (0.0074)
Market Beta	0.0036** (0.0016)	0.0042*** (0.0016)	0.0044*** (0.0016)	0.0021 (0.0021)	0.0027 (0.0020)	0.0034 (0.0020)
R-squared	0.264	0.264	0.264	0.271	0.271	0.271
Observations	26,967	26,967	26,967	26,967	26,967	26,967
Industry FE	No	No	No	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Sample excludes calendar years 2020 and 2021. The dependent variable is monthly excess return. “Dummy (top 20%)” equals 1 for firms in the top 20% of absolute emissions within the relevant scope (Scope 1, 2, or 3) in each cross-section. Columns (1)–(3) exclude industry fixed effects; (4)–(6) include industry fixed effects. CVIX denotes the Carbon Volatility Index (measured in index points). Controls are the same as in prior tables (momentum, book-to-market, leverage, CapEx/TA, PPE intensity, sales growth, EPS growth, return on equity, 1-year volatility, market beta). Two-way clustered standard errors are reported in parentheses. Significance:

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table VII. Excess Return Excluding Pandemic - Absolute

Monthly excess returns regressed on a high-emitter dummy (top 20% emissions, scope-specific) and its interaction with CVIX (sample excluding 2020–2021).

Variables	No Industry FE			With Industry FE		
	Scope 1	Scope 2	Scope 3	Scope 1	Scope 2	Scope 3
Intercept	0.0287*** (0.0041)	0.0285*** (0.0041)	0.0285*** (0.0040)	0.0473*** (0.0064)	0.0477*** (0.0062)	0.0490*** (0.0063)
Dummy (top 20%)	-0.0157*** (0.0050)	-0.0141*** (0.0055)	-0.0087* (0.0048)	-0.0170*** (0.0051)	-0.0117** (0.0058)	-0.0080 (0.0052)
Dummy $\times$ CVIX	0.0003*** (0.0001)	0.0003*** (0.0001)	0.0002* (0.0001)	0.0003*** (0.0001)	0.0003*** (0.0001)	0.0001* (0.0001)
Momentum	-0.0020 (0.0021)	-0.0016 (0.0020)	-0.0016 (0.0020)	-0.0083*** (0.0024)	-0.0082*** (0.0023)	-0.0081*** (0.0023)
Book-to-Market	-0.0157*** (0.0022)	-0.0153*** (0.0020)	-0.0153*** (0.0020)	-0.0198*** (0.0023)	-0.0198*** (0.0023)	-0.0197*** (0.0023)
Leverage	-0.0082*** (0.0030)	-0.0084*** (0.0030)	-0.0083*** (0.0030)	-0.0121*** (0.0043)	-0.0127*** (0.0043)	-0.0125*** (0.0043)
CapEx / TA	-0.0878*** (0.0211)	-0.0894*** (0.0212)	-0.0895*** (0.0213)	-0.1357*** (0.0288)	-0.1394*** (0.0287)	-0.1360*** (0.0284)
PPE intensity	0.0091*** (0.0031)	0.0114*** (0.0033)	0.0118*** (0.0032)	0.0066 (0.0051)	0.0056 (0.0052)	0.0065 (0.0051)
Sales growth	0.0016 (0.0117)	-0.0001 (0.0125)	-0.0001 (0.0124)	-0.0024 (0.0111)	-0.0026 (0.0120)	-0.0030 (0.0118)
EPS growth	0.0099** (0.0044)	0.0104** (0.0044)	0.0102** (0.0044)	0.0115** (0.0047)	0.0115** (0.0046)	0.0115** (0.0047)
Return on Equity	-0.0201*** (0.0039)	-0.0205*** (0.0039)	-0.0204*** (0.0039)	-0.0241*** (0.0047)	-0.0241*** (0.0046)	-0.0239*** (0.0047)
Volatility (1Y)	0.0214*** (0.0055)	0.0212*** (0.0055)	0.0216*** (0.0055)	0.0141* (0.0073)	0.0142* (0.0074)	0.0146** (0.0074)
Market Beta	0.0038** (0.0016)	0.0041*** (0.0016)	0.0042*** (0.0016)	0.0025 (0.0020)	0.0025 (0.0020)	0.0026 (0.0020)
R-squared	0.264	0.264	0.264	0.271	0.270	0.270
Observations	26,967	26,967	26,967	26,967	26,967	26,967
Industry FE	No	No	No	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Sample excludes calendar years 2020 and 2021. The dependent variable is monthly excess return. ‘Dummy (top 20%)’ equals 1 for firms in the top 20% of relative emissions (tons of CO₂-equivalent per thousand euros) within the relevant scope (Scope 1, 2, or 3) in each cross-section. Columns (1)–(3) exclude industry fixed effects; (4)–(6) include industry fixed effects. CVIX denotes the Carbon Volatility Index (measured in index points). Controls are the same as in prior tables (momentum, book-to-market, leverage, CapEx/TA, PPE intensity, sales growth, EPS growth, return on equity, 1-year volatility, market beta). Two-way clustered standard errors in parentheses. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table VIII. Excess Return Excluding Pandemic - Intensity

Monthly excess returns regressed on a high-emitter dummy (top 20% emissions intensity, scope-specific) and its interaction with CVIX (sample excluding 2020–2021).

Variables	No Firm FE			With Firm FE		
	Scope 1	Scope 2	Scope 3	Scope 1	Scope 2	Scope 3
Intercept	0.0309*** (0.0042)	0.0303*** (0.0042)	0.0301*** (0.0042)	0.0568*** (0.0062)	0.0564*** (0.0062)	0.0613*** (0.0071)
Dummy (top 20%)	-0.0117*** (0.0042)	-0.0164*** (0.0042)	-0.0180*** (0.0041)	-0.0051 (0.0057)	-0.0162*** (0.0054)	-0.0228*** (0.0052)
Dummy $\times$ CVIX	0.0003*** (0.0001)	0.0003*** (0.0001)	0.0003*** (0.0001)	0.0003*** (0.0001)	0.0003*** (0.0001)	0.0003*** (0.0001)
Momentum	-0.0025 (0.0019)	-0.0025 (0.0019)	-0.0026 (0.0019)	-0.0162*** (0.0026)	-0.0160*** (0.0026)	-0.0161*** (0.0026)
Book-to-Market	-0.0175*** (0.0021)	-0.0171*** (0.0020)	-0.0168*** (0.0020)	-0.0374*** (0.0035)	-0.0372*** (0.0035)	-0.0372*** (0.0035)
Leverage	-0.0101*** (0.0029)	-0.0100*** (0.0029)	-0.0101*** (0.0029)	-0.0100 (0.0077)	-0.0100 (0.0077)	-0.0102 (0.0077)
CapEx / TA	-0.0909*** (0.0199)	-0.0928*** (0.0198)	-0.0913*** (0.0199)	-0.0855** (0.0343)	-0.0851** (0.0340)	-0.0853** (0.0341)
PPE intensity	0.0119*** (0.0032)	0.0134*** (0.0032)	0.0137*** (0.0032)	-0.0068 (0.0110)	-0.0064 (0.0110)	-0.0065 (0.0111)
Sales growth	0.0152 (0.0138)	0.0142 (0.0138)	0.0130 (0.0137)	0.0085 (0.0143)	0.0084 (0.0143)	0.0080 (0.0142)
EPS growth	0.0070* (0.0037)	0.0072* (0.0038)	0.0068* (0.0038)	0.0080* (0.0043)	0.0083* (0.0044)	0.0076* (0.0044)
Return on Equity	-0.0175*** (0.0038)	-0.0177*** (0.0037)	-0.0179*** (0.0037)	-0.0292*** (0.0064)	-0.0290*** (0.0063)	-0.0291*** (0.0064)
Volatility (1Y)	0.0230*** (0.0052)	0.0222*** (0.0053)	0.0212*** (0.0052)	0.0215** (0.0085)	0.0208** (0.0085)	0.0215** (0.0085)
Market Beta	0.0020 (0.0016)	0.0025 (0.0016)	0.0029* (0.0016)	-0.0029 (0.0028)	-0.0027 (0.0028)	-0.0026 (0.0028)
R-squared	0.285	0.285	0.285	0.298	0.297	0.298
Observations	34,186	34,186	34,186	34,186	34,186	34,186
Firm FE	No	No	No	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The dependent variable is monthly excess return. “Dummy (top 20%)” equals 1 for firms in the top 20% of absolute emissions within the relevant scope (Scope 1, 2, or 3) in each cross-section. Columns (1)–(3) exclude firm fixed effects; (4)–(6) include firm fixed effects. CVIX denotes the Carbon Volatility Index. Controls are the same as in prior tables (momentum, book-to-market, leverage, CapEx/TA, PPE intensity, sales growth, EPS growth, return on equity, 1-year volatility, market beta). Two-way clustered standard errors are reported in parentheses. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table IX. Excess Return w Firm FEs - Absolute

Monthly excess returns regressed on a high-emitter dummy (top 20% emissions, scope-specific) and its interaction with CVIX with firm FEs.

Variables	No Firm FE			With Firm FE		
	Scope 1	Scope 2	Scope 3	Scope 1	Scope 2	Scope 3
Intercept	0.0309*** (0.0042)	0.0308*** (0.0042)	0.0306*** (0.0042)	0.0567*** (0.0062)	0.0571*** (0.0062)	0.0572*** (0.0062)
Scope Dummy	-0.0144*** (0.0051)	-0.0128** (0.0055)	-0.0113** (0.0045)	-0.0152*** (0.0057)	-0.0102 (0.0065)	-0.0079 (0.0052)
Dummy $\times$ CVIX	0.0003*** (0.0001)	0.0003** (0.0001)	0.0002** (0.0001)	0.0003*** (0.0001)	0.0003*** (0.0001)	0.0002** (0.0001)
Momentum	-0.0027 (0.0019)	-0.0024 (0.0019)	-0.0024 (0.0019)	-0.0161*** (0.0026)	-0.0160*** (0.0026)	-0.0159*** (0.0026)
Book-to-Market	-0.0175*** (0.0022)	-0.0170*** (0.0020)	-0.0170*** (0.0020)	-0.0373*** (0.0035)	-0.0372*** (0.0035)	-0.0372*** (0.0035)
Leverage	-0.0097*** (0.0029)	-0.0099*** (0.0029)	-0.0101*** (0.0029)	-0.0105 (0.0077)	-0.0100 (0.0077)	-0.0105 (0.0077)
CapEx / TA	-0.0897*** (0.0198)	-0.0916*** (0.0199)	-0.0907*** (0.0201)	-0.0843** (0.0343)	-0.0874** (0.0341)	-0.0857** (0.0341)
PPE intensity	0.0102*** (0.0031)	0.0125*** (0.0033)	0.0135*** (0.0032)	-0.0066 (0.0110)	-0.0061 (0.0109)	-0.0064 (0.0110)
Sales growth	0.0165 (0.0133)	0.0146 (0.0139)	0.0145 (0.0139)	0.0095 (0.0139)	0.0093 (0.0145)	0.0094 (0.0145)
EPS growth	0.0070* (0.0037)	0.0072* (0.0037)	0.0072* (0.0038)	0.0081* (0.0043)	0.0082* (0.0043)	0.0082* (0.0043)
Return on Equity	-0.0174*** (0.0038)	-0.0177*** (0.0037)	-0.0179*** (0.0037)	-0.0292*** (0.0064)	-0.0293*** (0.0063)	-0.0291*** (0.0064)
Volatility (1Y)	0.0219*** (0.0051)	0.0216*** (0.0051)	0.0218*** (0.0051)	0.0210** (0.0085)	0.0205** (0.0085)	0.0211** (0.0085)
Market Beta	0.0021 (0.0016)	0.0024 (0.0015)	0.0026* (0.0016)	-0.0029 (0.0028)	-0.0030 (0.0028)	-0.0029 (0.0028)
R-squared	0.285	0.285	0.285	0.298	0.297	0.297
Observations	34,186	34,186	34,186	34,186	34,186	34,186
Firm FE	No	No	No	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The dependent variable is monthly excess return. “Dummy (top 20%)” equals 1 for firms in the top 20% of relative emissions (tons of CO₂-equivalent per thousand euros) within the relevant scope (Scope 1, 2, or 3) in each cross-section. Columns (1)–(3) exclude firm fixed effects; (4)–(6) include firm fixed effects. CVIX denotes the Carbon Volatility Index. Controls are the same as in prior tables (momentum, book-to-market, leverage, CapEx/TA, PPE intensity, sales growth, EPS growth, return on equity, 1-year volatility, market beta). Two-way clustered standard errors are reported in parentheses. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table X. Excess Return w Firm FEs - Intensity

Monthly excess returns regressed on a high-emitter dummy (top 20% emissions intensity, scope-specific) and its interaction with CVIX with firm FEs.